

Worksheet 3: Section 1.3

Complex Numbers

Name: _____

Date: _____

Imaginary Numbers

$$i = \sqrt{-1} \implies i^2 = -1 \implies \sqrt{-b} = i\sqrt{b} \text{ for } b > 0$$

Standard Complex Form

$a + bi$ where a is the real part and b is the imaginary part

Adding and Subtracting Complex Numbers

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

Multiplying Complex Numbers

Use distributive property (FOIL), remember $i^2 = -1$.

Complex Conjugates

$$(a + bi)(a - bi) = a^2 + b^2 \quad (\text{always a real number})$$

Dividing Complex Numbers

Multiply numerator and denominator by the conjugate of the denominator.

$$\frac{a + bi}{c + di} \cdot \frac{c - di}{c - di}$$

Practice Problems

Show all work. Box your final answers.

Problem 1. Simplify each expression:

$a) \sqrt{-100}$

$b) \sqrt{-36}$

$c) \sqrt{-6}$

$d) \sqrt{-20}$

Solution:

Problem 2. Simplify each expression:

$a) \sqrt{-16} \cdot \sqrt{-25}$

$b) \sqrt{-40} \cdot \sqrt{-10}$

Solution:

Problem 3. Write in standard complex form $a + bi$:

$a) 3 + \sqrt{-16}$

$b) 5$

$c) 7i$

$d) \frac{6 + \sqrt{-8}}{2}$

Solution:

Problem 4. Fill in the pattern:

$$i^0 = \text{-----} \quad i^1 = \text{-----} \quad i^2 = \text{-----} \quad i^3 = \text{-----} \quad i^4 = \text{-----}$$

$$i^5 = \text{-----} \quad i^6 = \text{-----} \quad i^7 = \text{-----}$$

What pattern do you see? _____

Problem 4b. What is i^{23} ?

Solution:

Problem 5. Perform the operations:

$$(8 + 3i) + (5 - 2i)$$

$$(8 + 3i) - (5 - 2i)$$

Solution:

$$(5 - 3i) + 7i$$

$$(-5 - 2i) - (5 - 2i)$$

Solution:

Problem 6. Multiply:

$$-4i(3 - 2i)$$

$$(4 - 5i)(3 + 2i)$$

$$(6 - 7i)^2$$

Solution:

Problem 7. Multiply the conjugates:

$$(2-5i)(2+5i)$$

$$(4-3i)(4+3i)$$

$$4i \cdot (-4i)$$

Solution:

Problem 8. Divide:

$$\frac{4-3i}{4i}$$

Solution:

Problem 9. Divide:

$$\frac{4-3i}{5+2i}$$

Solution:

Answer Key

Answer Key

Problem 1: $10i, 6i, i\sqrt{6}, 2i\sqrt{5}$

Problem 2: $-20, -20$

Problem 3: $3 + 4i, 5 + 0i, 0 + 7i, 3 + i\sqrt{2}$

Problem 4: $i^0 = 1, i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i, i^6 = -1, i^7 = -i;$

Pattern: $1, i, -1, -i$ repeats every 4.

Problem 4b: $i^{23} = -i$

Problem 5: $13 + i, 3 + 5i, 5 + 4i, -10$

Problem 6: $-8 - 12i, 22 - 7i, -13 - 84i$

Problem 7: $29, 25, 16$

Problem 8: $-\frac{3}{4} - i$

Problem 9: $\frac{14}{29} - \frac{23}{29}i$